List Data Structure

In [76]:
```
l = [] #empty list
l
```

Out[76]:
```
[]
```

In [77]:
```
type(l)
```

Out[77]:
```
list
```

In [78]:
```
l.append(10)
l
```

Out[78]:
```
[10]
```

In [79]:
```
l
```

Out[79]:
```
[10]
```

In [80]:
```
l.append(10)
l.append(20)
l.append(30)
l.append(40)
l.append(50)
l
```

Out[80]: [10, 10, 20, 30, 40, 50]

In [81]:
```
len(l)
```

Out[81]: 6

In [82]:
```
l.append(10)
l
```

Out[82]: [10, 10, 20, 30, 40, 50, 10]

In [83]:
```
l1 = [10,2.3, 1+2j, True, 'nit',[1,2,3]]
l1
```

Out[83]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]

In [84]:
```
len(l1)
```

Out[84]: 6

In [85]:
```
print(l)
print(l1)
```

```
[10, 10, 20, 30, 40, 50, 10]
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [86]:
```
print(id(l))
print(id(l1))
```

```
3058669611904
3058670598400
```

In [87]:
```
l #l is called variable (10,20,30 are value or element)
```

Out[87]:
```
[10, 10, 20, 30, 40, 50, 10]
```

In [88]:
```
l2 = l.copy()
l2
```

Out[88]:
```
[10, 10, 20, 30, 40, 50, 10]
```

In [89]:
```
l2
```

Out[89]:
```
[10, 10, 20, 30, 40, 50, 10]
```

In [90]:
```
len(l2)
```

Out[90]:
```
7
```

In [91]:
```
l2.clear()
```

In [92]:
```
len(l2)
```

Out[92]:
```
0
```

In [93]:
```
l2
```

Out[93]:
```
[]
```

In [94]:
```
del l2
l2
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[94], line 2
      1 del l2
----> 2 l2

NameError: name 'l2' is not defined
```

In [95]:
```
l
```

Out[95]:
```
[10, 10, 20, 30, 40, 50, 10]
```

In [96]:
```
l.count(20)
```

Out[96]:
```
1
```

In [97]:
```
l.count(10)
```

Out[97]:
```
3
```

In [98]:
```
l.count(100)
```

Out[98]:
```
0
```

In [99]:
```
print(l)
print(l1)
```

```
[10, 10, 20, 30, 40, 50, 10]
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [100…

```
a= [1,2,3]
b= [4,5,6]

a.extend(b)
a
```

Out[100…

```
[1, 2, 3, 4, 5, 6]
```

In [101…

```
l2 = []
l2
```

Out[101…

```
[]
```

In [102…

```
l2.extend(l)
l2
```

Out[102…

```
[10, 10, 20, 30, 40, 50, 10]
```

In [103…

```
l2.extend(l1)
l2
```

Out[103…

```
[10, 10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [104…

```
print(l)
print(l1)
print(l2)
```

```
[10, 10, 20, 30, 40, 50, 10]
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
[10, 10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [105…

```
l[0]
```

Out[105…

```
10
```

In [107…

```
l[0]= 100 #list is muteable
l
```

Out[107…

```
[100, 10, 20, 30, 40, 50, 10]
```

In [109…

```
l[-2]= 500 #mutable
l
```

Out[109…

```
[100, 10, 20, 30, 40, 500, 10]
```

In [110…

```
l.index(100)
```

Out[110…

```
0
```

In [111…

```
l2
```

Out[111…

```
[10, 10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [112…

```
l2.index(10)
```

Out[112…

```
0
```

In [114…
```
l.insert(2, 25)
l
```

Out[114…
```
[100, 10, 25, 20, 30, 40, 500, 10]
```

In [115…
```
l2
```

Out[115…
```
[10, 10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [116…
```
l2.pop()
```

Out[116…
```
[1, 2, 3]
```

In [117…
```
l2
```

Out[117…
```
[10, 10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit']
```

In [118…
```
l2.pop()
```

Out[118…
```
'nit'
```

In [120…
```
l2.pop()
```

Out[120…
```
True
```

In [121…
```
l2
```

Out[121…
```
[10, 10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j)]
```

In [141…

```
l2.pop(2)
l2
```

Out[141…

```
[10, 10, 2.3]
```

In [142…

```
l2.remove(40)
l2
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[142], line 1
----> 1 l2.remove(40)
      2 l2

ValueError: list.remove(x): x not in list
```

In [136…

```
l2.pop(4)
```

Out[136…

```
(1+2j)
```

In [137…

```
l.reverse()
l
```

Out[137…

```
[500, 100, 40, 30, 25, 20, 10, 10]
```

In [138…

```
l.sort()
l
```

Out[138…

```
[10, 10, 20, 25, 30, 40, 100, 500]
```

In [139…
```
l
```
Out[139…
```
[10, 10, 20, 25, 30, 40, 100, 500]
```
In [140…
```
l.sort(reverse=True)
l
```
Out[140…
```
[500, 100, 40, 30, 25, 20, 10, 10]
```
In [143…
```
l3 = [10,20,[2,3],3]
```
In [144…
```
l3[2] = [2]
l3
```
Out[144…
```
[10, 20, [2], 3]
```
In [145…
```
l
```
Out[145…
```
[500, 100, 40, 30, 25, 20, 10, 10]
```
In [146…
```
30 in l
```
Out[146…
```
True
```
In [147…
```
500 in l
```
Out[147…
```
True
```
In [149…
```
5000 in l  #membership
```

Out[149…

```
False
```

In [150…

```python
for i in l:
    print(i)
```

```
500
100
40
30
25
20
10
10
```

In [ ]:

In [ ]:

In [ ]:

In [ ]: